# Toward Weak Signal Analysis in Hyperspectral Data: An Efficient Unmixing Perspective

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***Abstract*— Many unmixing methods hold the assumption that endmembers correspond to major land covers, but not true for some unmixing tasks where observed minor object signals corresponding to some special types of endmembers are relatively weak. When there exist weak signals that have low intensity potentially caused by subtle mixing abundance fractions regarding the endmembers of minor objects, the traditional unmixing techniques may fail. This article pioneers weak signal scenarios in hyperspectral unmixing using an efficient method called HyperWeak. In particular, HyperWeak involves a sparse nonnegative matrix factorization (NMF) model that contains two main parts, where the unsupervised part estimates the endmember and abundance matrices, and the supervised part ensures the minimal degradation of prior knowledge. To enhance the robustness of the HyperWeak model, this article considers a reweighted sparsity constraint to boost the sparseness of the abundance matrix. For effectively solving optimization problems, a Nesterov's optimal gradient method (OGM) is used in this article. Experiments conducted on synthetic and real hyperspectral images indicate that HyperWeak can improve the unmixing performances of hyperspectral data in weak signal situations.**

***Index Terms*— Hyperspectral image, hyperspectral unmixing, nonnegative matrix factorization (NMF), weak signal analysis.**

## I. Introduction

HYPERSPECTRAL remote sensing technology, also known as imaging spectroscopy, records the radiation reflected or emitted from the surface in tens or hundreds of continuously spaced narrow spectral bands in each pixel [1], which has been well applied to different fields, including heavy metal pollution [2], precision agriculture [3], mineral exploration [4], water monitoring [5], and so on. Due to the limited spatial resolution of the hyperspectral sensor or mixed homogeneous area of distinct materials in hyperspectral images, the pixel tends to be linearly mixed by several distinct pure substances, i.e., *endmembers*, with corresponding fractions, i.e., *abundances* [6], [7]. Endmembers normally correspond to familiar macroscopic objects in the scene, such as water, soil, metal, vegetation, and so on [6]. The process of decomposing mixed pixels into endmember and abundance matrices is called hyperspectral unmixing [8], [9].

In the nonlinear mixing model, it assumes that the light follows multiple scattering phenomena due primarily to physical interactions among the light scattered by multiple materials in the scene, implying that establishing a more complex physical model to simulate potential light scattering types is highly demanded [10]. The discussion of interaction types leads to some representative works, including [11]–[14]. Many attempts, however, are devoted to the linear mixing mode due mainly to the fact that it is an acceptable approximation of the light scattering mechanisms in many real scenarios and is the basis of a plethora of unmixing models in the past several decades despite its simplicity [8]. The process of first identifying endmembers and then estimating corresponding abundance fractions forms a two-step-based linear unmixing direction. This direction focuses mainly on two independent aspects of extracting endmembers over an already estimated number of endmembers and estimating abundance fractions using a fully constrained least-square method [15]–[23]. Statistical-based algorithms give more importance to nonnegative matrix factorization (NMF)-based methods, which formulate a linear unmixing problem as the process of simultaneously estimating endmember and abundance matrices using multiplicative update rules [7], [24]–[29]. Sparse regression-based linear unmixing models introduce an independent spectral library and only yield corresponding abundance fractions [30]–[34]. The abovementioned strategies form the main lines of traditional unmixing research.

Traditional unmixing strategies, however, assume that endmembers correspond to representative land covers, but not true for some unmixing tasks where observed minor object signals corresponding to some special types of endmembers are relatively weak. In particular, many unmixing methods extract/generate endmembers by assuming they occupy the vertices of the data-enclosing simplex, and all mixed pixels occupy the interior of the simplex volume. When weak signals exist, macroscopic land covers still dominate information components, and weak signals only contribute to extremely limited information, as they are some mixed pixels with subtle mixing abundance fractions regarding the endmembers of minor objects (or called weak endmembers). Normally, the weak signal is a multidisciplinary concept that the useful information is hidden in the background [35], which has been

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concerned in other fields, such as communications [36], but not yet received high attention in the remote sensing field. When observed hyperspectral images occur to weak signals, the process of identifying corresponding endmembers and unmixing abundance fractions of weak signals is difficult.

Although the effect of weak signals on the analysis of the background is sometimes negligible [37], the weak signals still deserve further exploitation and analysis, because, in most cases, they indicate some small but useful information, such as pollution or minerals. A representative example of water quality monitoring exists in algal bloom detection that involves indirectly measuring chlorophyll concentration. The chlorophyll, which is a pigment existing in green vegetation, absorbs visible light and water–air interfaces, resulting in high reflectance in the near-infrared wavelengths [38]. But, a low chlorophyll concentration of $<2$ mg/m$^3$ only has higher reflectances in the blue part of the spectrum (400–500 nm) [39]. Another example can be found from the most widely used normalized difference vegetation index (NDVI), which is a dimensionless index that describes the difference between visible and near-infrared reflectance of vegetation cover and can be used to estimate the density of green on an area of land [40]. But, NDVI tends to saturate, as the vegetation cover is *dense* [38]. That is, if the vegetation coverage, as well as low chlorophyll concentrations, is not saturation enough compared with the background covers, the interpretation accuracy for the observed ground cannot meet requirements [38], [41]. In these cases, traditional unmixing strategies tend to meet the bottleneck due to the weak signal situations in real applications.

For the weak signal, the problem of solving this case may relate sorely to two points. The first concern with the issue is that the initialization and optimization may be difficult, as some targets have subtle fractions. If the ground objects exist with subtle fractions, their spectral curves barely describe distinct features but may have high correlations to other mixture substances. For instance, when the river water contains iron (Fe), the spectrum curve only has changes in the absorption region 800–1000 nm [42], implying that the spectral correlations between two objects are similar. Therefore, the initialized signatures corresponding to weak signals will show high correlations to those of substances that occupy major mixing fractions. The second concerns the fact that the weak signals are normally hidden in the spectrally mixed background, and hence, a detailed knowledge of water quality parameters and spectral information is required [43]. When *a priori* of knowledge is introduced to detect and separate the corresponding weak signal, it may degrade during the optimization task.

Recently, semi-supervised NMF-based methods are proposed to unmix hyperspectral data by introducing an endmember library [44]–[47]. Nevertheless, they hold a similar belief that an area has certain land covers, such as grass or water, whose spectral signatures can be retrieved from a standard library or by ground survey [44]. When some of the pure endmembers are known and available prior to conducting unmixing tasks, they will be treated as an endmember library and subsequently replace the initialized endmembers [45]. However, there still exist potential quandaries. First, the introduced endmember library may mismatch already initialized endmembers, if the spectral correlations between them are low, because initialized endmembers are normally not pure and corrupted by noises. Second, if the endmember library is not introduced to the semi-supervised NMF, the initialized endmembers still have the ability in helping NMF accurately estimate endmember and abundance even though they are not pure enough [48]. The main reason is that the optimization process tends to optimize initialized non-pure endmembers as the final pure signatures, so that each optimized endmember can reside at the vertex of the data-enclosing simplex.

This article proposes an efficient approach to solve weak signal scenarios in hyperspectral data, called HyperWeak. The proposed HyperWeak is developed based on sparse NMF with two fidelity terms. The first unsupervisedly estimates abundance fractions and endmembers, where the initial endmember matrix is augmented by initialized endmembers and prior knowledge. The second supervisedly minimizes the degradation of prior knowledge related to weak signals during the optimization process. A reweighted sparsity constraint and a robust orthogonal subspace projection (OSP) method, called Robust-OSP, are proposed to enhance the unmixing performances of HyperWeak. The Nesterov's optimal gradient method (OGM) is applied to HyperWeak for fast convergence purposes.

We make two chief contributions.

1) This article introduces a weak signal scenario that assumes some observed targets have low intensity potentially caused by subtle mixing abundance fractions. To solve this issue, this article proposes HyperWeak, which involves two main parts. The first part unsupervisedly minimizes the errors of image reconstruction, and the second part supervisedly minimizes the degradation of prior knowledge. Besides, a reweighted sparsity constraint combined with a denoising process is applied to HyperWeak.
2) This article proposes a new endmember initialization algorithm named Robust-OSP. Robust-OSP first transforms hyperspectral data into a subspace and then successively finds endmembers according to the maximum projections. Compared with vertex component analysis (VCA), Robust-OSP can produce stable and repeatable initialization results.

This article is divided into five sections, which can be given as follows. Section II reviews related preliminaries. The HyperWeak is introduced in Section III in detail. Section IV displays experimental results. Section V relates the findings and observations to other relevant studies. Section VI concludes with some remarks.

## II. Preliminaries

### A. Linear Mixing Model

A linear mixing model considers that light follows a linear combination of different materials (or endmembers). Suppose $Y \in \mathbb{R}^{L\times N}$ is a hyperspectral image with $L$ bands and $N$ total pixels. It can be formulated by considering endmember,

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abundance, and noise matrices

$$Y = WH + G \tag{1}$$

where $W \in \mathbb{R}^{L\times p}$, $H \in \mathbb{R}^{p\times N}$, and $G \in \mathbb{R}^{L\times N}$ are endmember, abundance, and Gaussian additive noise matrices, respectively, and $p$ is the number of endmembers, which can be estimated by virtual dimensionality [49].

### *B. Nonegative Matrix Factorization*

NMF-based hyperspectral unmixing algorithms involve jointly estimating endmember and abundance matrices from hyperspectral data. Based on the previous notations, NMF can be expressed as

$$Y \approx WH \tag{2}$$

where $W \succeq 0$ and $H \succeq 0$. To measure the difference between the original signal $Y$ and the reconstructed signal $WH$, its cost function is considered as

$$\begin{aligned} &\min_{W,H} \frac{1}{2}\|Y - WH\|_F^2 \\ &\text{s.t. } W \succeq 0, \quad H \succeq 0, \ \mathbf{1}_p^T H = \mathbf{1}_N^T \end{aligned} \tag{3}$$

where $\|\cdot\|_F^2$ denotes the Frobenius norm. The multiplied iterative optimization is commonly used to solve (3), and the multiplicative update rules for two variables in the $i$th iteration are given as follows:

$$\begin{aligned} W^{(i+1)} &\leftarrow W^{(i)} \otimes \left[Y\left(H^{(i)}\right)^T\right] \oslash \left[WH^{(i)}\left(H^{(i)}\right)^T\right] \\ H^{(i+1)} &\leftarrow H^{(i)} \otimes \left[\left(\tilde{W}^{(i+1)}\right)^T \tilde{Y}\right] \oslash \left[\left(\tilde{W}^{(i+1)}\right)^T \tilde{W}^{(i+1)} H^{(i)}\right] \end{aligned} \tag{4}$$

where $\otimes$ and $\oslash$, respectively, denote element-wise multiplication and element-wise division. Both $\tilde{Y}$ and $\tilde{W}$ represent augmented matrices for reinforcing an abundance sum-to-one constraint, and the augmented matrices can be given by

$$\begin{aligned} \tilde{W} &\leftarrow \left[W; \delta \mathbf{1}_p^T\right] \\ \tilde{Y} &\leftarrow \left[Y; \delta \mathbf{1}_N^T\right] \end{aligned} \tag{5}$$

where $\delta$ denotes the parameter of abundance sum-to-one constraint.

## III. Hyperspectral Weak Signal Analysis Method With a Sparse NMF

### *A. Problem Formulation and Motivation*

A hyperspectral imaging sensor captures a determined area of the Earth's land surface in which the so-called endmembers correspond to familiar macroscopic objects in the observed scene, such as water, soil, metal, vegetation, and so on. This imaging process equally scans some minor objects, such as contaminated water, small ground man-made targets, rare minerals, and so on. In real applications, for example, in water quality monitoring, the minor pure substances may be factory sewage, sand, algae, and so on, that exist in the pure water body in a mixture manner with subtle mixing fractions. In this case, the identification and extraction of minor substances in the observed scene are sometimes very important that require a more efficient analysis paradigm. Due mainly to subtle mixing fractions, however, minor substances are not easily extracted, which forms the main motivation of this study that exploits the importance of prior knowledge for the purpose of unmixing hyperspectral data under weak signal situations.

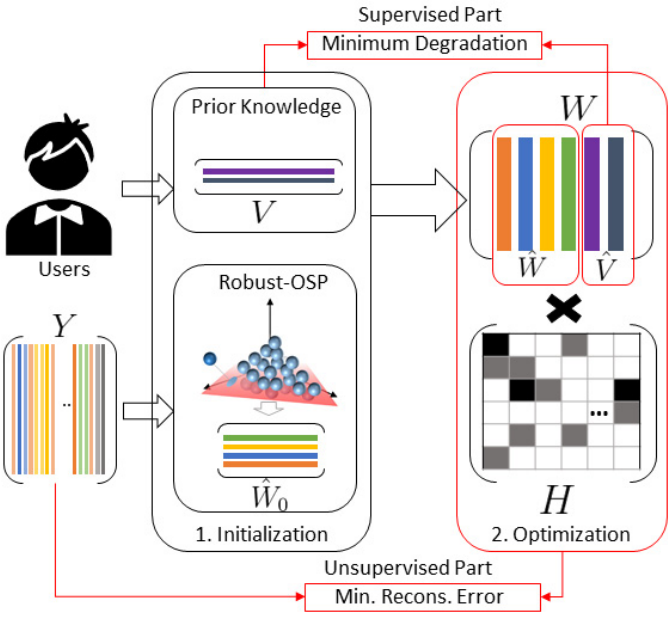



Fig. 1. Visual explanation of HyperWeak. In the initialization procedure, HyperWeak uses Robust-OSP to extract endmembers $\hat{W}_0$ related to major land covers and simultaneously supplies prior knowledge $V$ corresponding to weak endmembers. In the optimization procedure, HyperWeak unsupervisedly optimizes endmembers $[\hat{W}, \hat{V}]$ and corresponding abundance fractions $H$ and supervisedly minimizes the deterioration of prior knowledge.

### *B. Model*

The traditional NMF is required to jointly estimate endmember and abundance matrices over an already determined number of endmembers and initialization inputs. However, when some objects are weak, the initialization and optimization for those objects are difficult.

Therefore, we propose the HyperWeak method that contains two parts. The unsupervised part of HyperWeak focuses on estimating endmember and abundance matrices, where the endmembers contain Robust-OSP-initialized endmembers and prior knowledge. The supervised part ensures minimal degradation of prior knowledge that is believed to potentially exist in a scene. It is worth mentioning that, in real applications, two major approaches can be used to obtain prior knowledge. First, users can directly select specific prior knowledge from the spectral library to observe whether the image contains target weak signals. Second, users can apply image-based analysis and ground survey processes to obtain prior knowledge. For example, if the river is contaminated by algae or chemical substances, users can first analyze observed hyperspectral data to estimate the potential regions containing weak signals. Then, they can take water samples and conduct laboratory-based measurements for observing the true type and spectral signature of weak signals. Similar procedures can be found from other representative works [50], [51]. Fig. 1 visually

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explains the flowchart of HyperWeak. The unsupervised model can be formulated by

$$\min_{\hat{W},\hat{V},H} \frac{1}{2}\left\|Y-\left[\hat{W},\hat{V}\right]H\right\|_F^2 \quad \text{s.t. } \hat{W}\succeq 0,\quad \hat{V}\succeq 0, H\succeq 0,\ \mathbf{1}_{p+q}^T H=\mathbf{1}_N^T \tag{6}$$

where $\hat{W}\in\mathbb{R}^{L\times p}$ represents normal endmembers corresponding to familiar macroscopic land covers, and it can be unsupervisedly initialized using Robust-OSP. $\hat{V}\in\mathbb{R}^{L\times q}$ denotes weak endmembers corresponding to minor objects. It is worth mentioning that both $\hat{W}$ and $\hat{V}$ will be optimized, which may heavily affect prior knowledge. To solve this issue, the supervised part is considered to minimize the degradation of $\hat{V}$, given as

$$\min_{\hat{V}} \frac{1}{2}\left\|V-\hat{V}\right\|_F^2 \quad \text{s.t. } \hat{V}\succeq 0 \tag{7}$$

where $V$ stands for prior knowledge to ensure minimal deterioration of $\hat{V}$.

By joining (6) and (7) and simultaneously incorporating a sparsity constraint, we propose the HyperWeak framework, expressed as

$$\min_{W,H} \frac{1}{2}\underbrace{\|Y-WH\|_F^2}_{\text{Unsupervised}}+\frac{\tau}{2}\underbrace{\|V-WA\|_F^2}_{\text{Supervised}}+\lambda\underbrace{\|R\otimes H\|_1}_{\text{Regularization}} \quad \text{s.t. } W\succeq 0,\quad H\succeq 0,\ \mathbf{1}_{p+q}^T H=\mathbf{1}_N^T \tag{8}$$

where the first and second parts are fidelity terms, respectively, used to minimize the error of image reconstruction and degradation of *a priori* of knowledge, and the third term denotes a reweighted sparsity regularizer. Besides, $W=[\hat{W},\hat{V}]$ denotes an augmented endmember matrix, and $A=[0_{p\times q};I_{q\times q}]$ represents an auxiliary matrix indicating the position of $\hat{V}$. $\tau$ and $\lambda$ equally represent two parameters. $R$ denotes a reweighting matrix used for enhancing the sparseness of abundance matrix $H$ [52].

On the one side, $R$ is normally designed, as a matrix $1\oslash(|H|+\text{eps})$ with each element is often in inverse proportion to that of $H$ aiming at penalizing the nonzero values on the solution. On the other side, each abundance map intrinsically contains local similarities between two neighboring pixels, but the independent spatial exploitation, such as total variation regularization, requires more computational time. Recently, Zhao *et al.* [53] proposed a plugged prior regularization that minimizes the difference between optimized abundance fractions and denoised abundance fractions. Inspired by Zhao's interesting work, here, we introduce a much simpler strategy during the processing of reweighting. We first denoise the abundance matrix using filtering, such as mean value filtering, Wiener filtering, and so on, and then generate $R$ based on the filtered $H$. The processing of generating the reweighting matrix $R$ in the $i$th iteration can be given by

$$R^{(i)}=1\oslash\left(\left|\mathcal{F}\left(H^{(i-1)}\right)\right|+\text{eps}\right) \tag{9}$$

where $\mathcal{F}(\cdot)$ denotes a filtering operator, and eps is the predetermined positive constant.

### C. Optimization

Taking the subproblem of $H$ as an example, this section provides its optimization details based on Nesterov's OGM. As $F(W^{(i)},H)$ is convex under the $i$th iteration and $\nabla_H F(W^{(i)},H)$ is Lipschitz continuous, OGM alternatively solves model (8) by

$$H_k=\arg\min_{H\geq 0} F\left(W^{(i)},S_k\right)+\left\langle\nabla_H F\left(W^{(i)},H\right),H-S_k\right\rangle+\frac{LC}{2}\|H-S_k\|_F^2 \tag{10}$$

and

$$S_k=H_k+\frac{\alpha_k-1}{\alpha_{k+1}}(H_k-H_{k-1}) \tag{11}$$

where $LC$ is the Lipschitz constant, $H_k$ contains the approximate solution by minimizing (10) over $H$, and $S_k$ stores the search point constructed by linearly combining the latest two approximate solutions. The combination coefficient $\alpha_k$ is updated in the $k$th iteration by

$$\alpha_{k+1}=\frac{1+\sqrt{4\alpha_k^2+1}}{2}. \tag{12}$$

For the abundance sum-to-one and nonnegative constraints, we, respectively, consider the augmented matrix strategy using (5) for sum-to-one constraint and a projected operator for nonnegative constraint by

$$H_k=\mathcal{P}\left(S_k-\frac{1}{LC}\nabla_H F\left(W^{(i)},S_k\right)\right) \tag{13}$$

where $\mathcal{P}(\cdot)$ denotes the operator that projects negative values into zeros.

The $i$th iteration of $H$ can be expressed in Algorithm 1. The stopping criterion of Algorithm 1 considers the following method:

$$\left\|\nabla_H^P F\left(W^{(i)},H_k\right)\right\|_F \leq \max\left(10^{-3},\epsilon_H\right)\times\left\|\left[\nabla_H^P F\left(W^{(1)},H^{(1)}\right),\nabla_H^P F\left(W^{(1)},H^{(1)}\right)^T\right]\right\|_F \tag{14}$$

where $\epsilon_H$ is a tolerance for $H$ optimization subproblem. $\nabla_H^P F(W^{(i)},H_k)$ is a projected gradient defined by

$$\nabla_H^P F\left(W^{(i)},H_k\right)=\begin{cases}\nabla_H F\left(W^{(i)},H_k\right)_{ij}, & \text{if } (H_k)_{ij}>0\\ \min\left\{0,\nabla_H F\left(W^{(i)},H_k\right)_{ij}\right\}, & \text{if } (H_k)_{ij}=0.\end{cases} \tag{15}$$

By alternatively optimizing two subproblem with respect to $W$ and $H$ according to Algorithm 1 in each iteration, we can clearly implement the details of HyperWeak by Algorithm 2. Two general stop criteria are used for Algorithm 2. The first considers the number of iterations. The second considers the convergence residuals defined by $O^{(i-1)}-O^{(i)}$, where $O^{(i)}=\|Y-W^{(i)}H^{(i)}\|_F^2$ and $i\geq 2$. If the residuals $O^{(i-1)}-O^{(i)}$ meet a predefined threshold, the optimization process will stop.

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**Algorithm 1** OGM

**Require:** $W^{(i)}$,$H^{(i)}$;
**Ensure:** optimal $H^{(i+1)}$
Initialize $H_0 = H^{(i)}$, $S_0 = H^{(i)}$, $\alpha_0 = 1$, $LC = \left\| \left(W^{(i)}\right)^T W^{(i)} \right\|_2$, $k = 0$;
**repeat**
  Update $H_k$ via $H_k = \mathcal{P}\left(S_k - \frac{1}{LC} \nabla_H F\left(W^{(i)}, S_k\right)\right)$;
  Update $\alpha_{k+1}$ via $\alpha_{k+1} = \frac{1+\sqrt{4\alpha_k^2+1}}{2}$;
  Updat $S_{k+1}$ via $S_k = H_k + \frac{\alpha_k - 1}{\alpha_{k+1}}(H_k - H_{k-1})$;
  $k \leftarrow k + 1$;
**until** Stopping criterion is satisfied
$H^{(i+1)} \leftarrow H_k$;

**Algorithm 2** HyperWeak

**Require:** $Y$, $V$, $p$, $q$, $\lambda$, $\tau$, $\delta$, $T_{max}$, $Tol$;
**Ensure:** optimal $W$, $H$
Initialize $\hat{W}_0$ using Robust-OSP over estimated $p$ and initialize $\hat{W}$ using $\hat{W}_0$; Initialize number of endmembers $p+q$; Initialize $\hat{V}$ using $V$; Initialize $W^{(0)}$ by combining $\hat{W}$ and $\hat{V}$; Initialize $H^{(0)}$ using fully constrained least-square over $W^{(0)}$; Initialize $A$; Initialize $i = 0$;
**repeat**
  $W^{(i+1)} \leftarrow OGM\left(W^{(i)}, H^{(i)}, \nabla_W F\left(W, H^{(i)}\right)\right)$;
  $\tilde{W}^{(i+1)} \leftarrow \left[W^{i+1}; \delta \mathbf{1}_{p+q}^T\right]$;
  $\tilde{Y} \leftarrow \left[Y; \delta \mathbf{1}_N^T\right]$;
  $R^{(i)} \leftarrow 1 \oslash \left(\left|\mathcal{F}\left(H^{(i)}\right)\right| + \text{eps}\right)$;
  $H^{(i+1)} \leftarrow OGM\left(\tilde{W}^{(i+1)}, H^{(i)}, \nabla_H F\left(\tilde{W}^{(i+1)}, H\right)\right)$;
  $i \leftarrow i + 1$;
**until** Stopping criteria are satisfied

### *D. Initialization*

The most important step prior to conducting NMF-based methods lies in the problem of initializing endmember and abundance matrices. When NMF-based methods meet a poor initialization process, they are more likely stuck in the local minima, so that the unmixing performances extensively degrade. The current publications in hyperspectral unmixing give mainly importance to VCA [48] or spectral information divergence [54] for generating initial endmembers and subsequently producing initial abundance fractions according to already determined endmembers. However, the algorithmic structure of such methods involves a random process, which makes the repeatable initialization nearly impossible. This article equally puts emphasis on the initialization process particularly for the initial endmember matrix and proposes a robust OSP method based on OSP [19]. OSP successively selects the endmember by projecting entire spectral vectors onto a direction orthogonal to the subspace spanned by the already determined endmembers. Unlike OSP, Robust-OSP first transforms hyperspectral data into $p$-dimensional subspace. Then, $p$ endmembers can be found from dimensionality reduced subspace by identifying the maximal projections. Finally, after obtaining $p$ endmembers from dimensionality reduced subspace, they will be inversely transformed into the original domain. Compared with OSP, Robust-OSP ensures lower computational complexity and provides noise-reduced endmembers.

**Algorithm 3** Initialization

**Require:** $Y$, $p$;
**Ensure:** $W_0$, $H_0$
$U = \text{eigs}(YY^T/N)$;
$X = U^T Y$; *%Subspace Transformation*
$X_{i,j} \leftarrow X_{i,j}/x^T \upsilon$;
$E_1 = \arg\max_{x \in X} x^T x$;
**for** $i \rightarrow p - 1$ **do**
  $\mathcal{S}_{E_i}^{\perp} = I - E_i \left(E_i^T E_i\right)^{-1} E_i^T$;
  $e_{i+1} = \arg\max_{x \in X} \mathcal{S}_{E_i}^{\perp} x$;
  $E_{i+1} = \left[E_i, e_{i+1}\right]$;
**end for**
$W_0 = UE$; *%Invese Transformation*
$H_0 = \text{fcls}(Y, W_0)$;

Let $Y$ denote a hyperspectral matrix with $L$ bands and $N$ observed samples. Robust-OSP first generates its sample correlation matrix determined by calculating $YY^T/N$ and finds left eigenvector matrix $U$. Then, Robust-OSP projects entire samples into $U$ for producing low-dimensional space $X = U^T Y$. To avoid displacements between the original data and projected ones, that a scaling factor $\upsilon = (1/p)\sum_{i=1}^{p} U_i$ is introduced, such that $x^T \upsilon > 0$, where $U_i$ denotes the $i$th column of $U$. A similar strategy is adopted in previous publications, including [20], [55], and [56].

The first endmember in $X$ is defined by

$$e_1 = \arg\max_{x \in X} x^T x \tag{16}$$

and the other endmembers are successively searched by projecting data onto a subspace spanned by already determined endmembers $E_{p-1} = [e_1, e_2, \ldots, e_{p-1}]$, expressed as

$$e_p = \arg\max_{x \in X} \mathcal{S}_{E_{p-1}}^{\perp} x \tag{17}$$

where $e_p$, $\mathcal{S}_{E_{p-1}}^{\perp} = I - E_{p-1}(E_{p-1}^T E_{p-1})^{-1} E_{p-1}^T$, and $I$, respectively, denote the $p$th endmember, the orthogonal subspace projector, and the identity matrix. The final $L$-dimensional initial matrix can be transformed by $W_0 = UE$, and the initial abundance matrix $H_0$ can be estimated by least-square-based methods, such as fully constrained least-square.

### *E. Complexity*

The main cost time of HyperWeak is spent on the calculation of the gradients $\nabla_W F(W, H)$ and $\nabla_H F(W, H)$, where $\nabla_W F(W, H) = W(HH^T + \tau AA^T) - (\tau VA^T + YH^T)$ and $\nabla_H F(W, H) = W^T WH + (\lambda R - W^T Y)$. For the first term of $\nabla_W F(W, H)$, its computational complexity is $\mathcal{O}(N\xi^2 + \xi^4 q) + T \times \mathcal{O}(L\xi^2)$, in which $T$ denotes the number of iterations and $\xi = p + q$, and the computational cost of the second term is $\mathcal{O}(LN\xi + L^2\xi^2 q)$. For $\nabla_H F(W, H)$, its computational cost is $\mathcal{O}(L\xi^2 + LN\xi + N\xi) + T \times \mathcal{O}(N\xi^2)$. The overall computational cost of HyperWeak is $\mathcal{O}(L^2\xi^2 q + N\xi^2 + L\xi^2 + 2LN\xi + +\xi^4 q) + T \times \mathcal{O}(L\xi^2 + N\xi^2)$.

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TABLE I
COMPUTATIONAL COMPLEXITY AMONG ROBUST-OSP, OSP, AND VCA

| Algorithms | Complexity (flops) |
|---|---|
| Robust-OSP | $\mathcal{O}\left(p^3N\right)$ |
| OSP | $\mathcal{O}\left(pL^2N\right)$ |
| VCA | $\mathcal{O}\left(2p^2N\right)$ |

For Robust-OSP, it consists of three main steps. The first step transforms $Y \in \mathbb{R}^{L\times N}$ into low-dimensional data $X \in \mathbb{R}^{p\times N}$ using singular values decomposition, which has negligible computational cost in the MATLAB platform. The second step finds the low dimensional endmembers with complexity $\mathcal{O}(p^3N + p^2N)$. The third step generates orthogonal subspace projectors with total complexity $\mathcal{O}(\sum_{r=1}^{p-1} p^2r + pr^2 + 3r^3)$, where $r$ represents the number of already determined endmembers. Since $N \gg p$, the overall complexity of Robust-OSP is $\mathcal{O}(p^3N)$. Here, we list the computational complexity of some representative endmember extraction algorithms for comparison, including VCA and OSP (see Table I). According to the complexity of the three methods, we found that the Robust-OSP has a lower computational cost than that of OSP, because $p^2$ is much less than $L^2$. We are also aware that compared with VCA, the Robust-OSP still requires more computational cost owing to its three-order form. However, an obvious advantage is that Robust-OSP has stable initialization performances, which may improve the applicability of Robust-OSP.

## IV. RESULTS

### A. Experimental Setup

*1) Benchmark Methods:* Since the proposed HyperWeak is established based on NMF, we consider six NMF-based benchmark unmixing methods, including spatially group sparsity constrained NMF (SGS-NMF) [28], total variation regularized reweighted sparse NMF (TV-RSNMF) [29], sparse deep NMF with total variation (SDNMF-TV) [57], manifold regularized sparse NMF (GL-NMF) [27], $L_{1/2}$-NMF [26], and minimum volume simplex constrained NMF (MVC-NMF) [7] to validate the experimental performances.

Note that, for fair experimental comparison purposes, six competitive methods also use prior knowledge of weak endmembers in the initialization step. That is to say, HyperWeak and six comparison methods have the same initialization inputs. All the algorithms ran on a PC with an Intel Core i7-10700 at 2.90 GHz and 16-GB RAM.

*2) Evaluation Metrics:* Two benchmark metrics, i.e., spectral angle distance (SAD) and root mean square error (RMSE), are used to assess the experimental performances of different unmixing methods. The former verifies the spectral similarity between two signatures, which can be expressed as

$$\mathrm{SAD}_k = \arccos\left(\frac{\hat{s}_k^T s_k}{\|\hat{s}_k\|\|s_k\|}\right) \tag{18}$$

where $\hat{s}_k$ and $s_k$ represent two signatures. Low SAD values mean high similarity between two signatures. The latter evaluates the error between two abundance maps, given by

$$\mathrm{RMSE}_k = \left(\frac{1}{N}\|\hat{a}_k - a_k\|_2^2\right)^{1/2} \tag{19}$$

where $\hat{a}_k$ and $a_k$ denote abundance maps. The lower RMSE values the better estimation performances.

TABLE II
PARAMETER SETTINGS IN DIFFERENT EXPERIMENTAL SCENARIOS

| | Size | SNR(dB) | no. of endm. | no. of weak endms | max purity of normal endm. | max purity of weak endms |
|---|---|---|---|---|---|---|
| Exp. 1 | 58*58*224 | 30 | 6 | 1 | 0.8 | —[a] |
| Exp. 2 | 58*58*224 | 30 | 6 | — | 0.8 | 0.15 |
| Exp. 3 | 58*58*224 | — | 6 | 1 | 0.8 | 0.15 |
| Exp. 4 | 58*58*224 | 30 | 6 | 1 | 0.8 | 0.15 |
| Exp. 5 | 100*100*198 | /[b] | 5 | 1 | 1 | 0.10 |
| Exp. 6 | 100*100*198 | / | 6 | 2 | 1 | 0.10, 0.10 |
| Exp. 7 | 250*191*183 | / | 12 | 1 | / | / |
| Exp. 8 | 58*58*224 | 30 | 6 | 1 | 0.8 | 0.15 |
| Exp. 9 | 58*58*224 | 30 | 6 | 1 | 0.8 | 0.15 |
| Exp. 10 | 58*58*224 | 30 | 6 | 1 | 0.8 | 0.15 |
| Exp. 11 | 58*58*224 | 30 | 6 | 1 | 0.8 | 0.15 |

[a] "—" denotes that the experimental setting is an experimental variable.
[b] "/" denotes that the experimental setting is not available.

*3) Hyperspectral Synthetic Datasets:* In this article, different synthetic datasets based on various parameter settings are used for experimental validation purposes. Assume there is a desired image that contains $nr \times nc$ pixels with $p$ endmembers selected from the U.S. Geological Survey (USGS) library, where $nr$ and $nc$, respectively, represent the width and the height. Following the linear mixing model and the method proposed in [7], we can generate the desired image with plenty of mixed pixels produced by a low-pass filter with maximum abundance fixed at 0.8.

To simulate a weak signal scenario in synthetic images, we select $q(q < p)$ signatures and reduce their abundance to a predetermined threshold. Concretely, suppose the number of weak endmembers is $q$, and the maximum abundance fractions related to weak endmembers are fixed at $\epsilon$. We first initialize an abundance map sized $nr \times nc$ with full-zero elements. Then, we select an abundance map from generated abundance maps and find the indices of the lowest values from the determined map. Third, the abundance map corresponding to the weak endmember is updated by $\epsilon$ on the determined indices. Fourth, the abundance map of the normal signal is updated by $(1-\epsilon)/p$ on the determined indices. Fifth, we update the lowest values of abundance map corresponding normal signal by $(1-\epsilon)/p$. By iteratively performing the steps mentioned earlier for $q$ weak endmembers, we can obtain the abundance maps that contain weak signals under the linear mixing assumption.

Zero mean Gaussian noises are added to the synthetic images with different signal-to-noise ratios (SNRs) for the purpose of simulating real situations. The detailed experiment settings are displayed in Table II.

*4) Hyperspectral Semi-Synthetic Real Datasets:* Since endmember and abundance matrices of Jasper Ridge are available online, we design a strategy to generate desired semi-synthetic real datasets by supplying several weak objects to real hyperspectral data.

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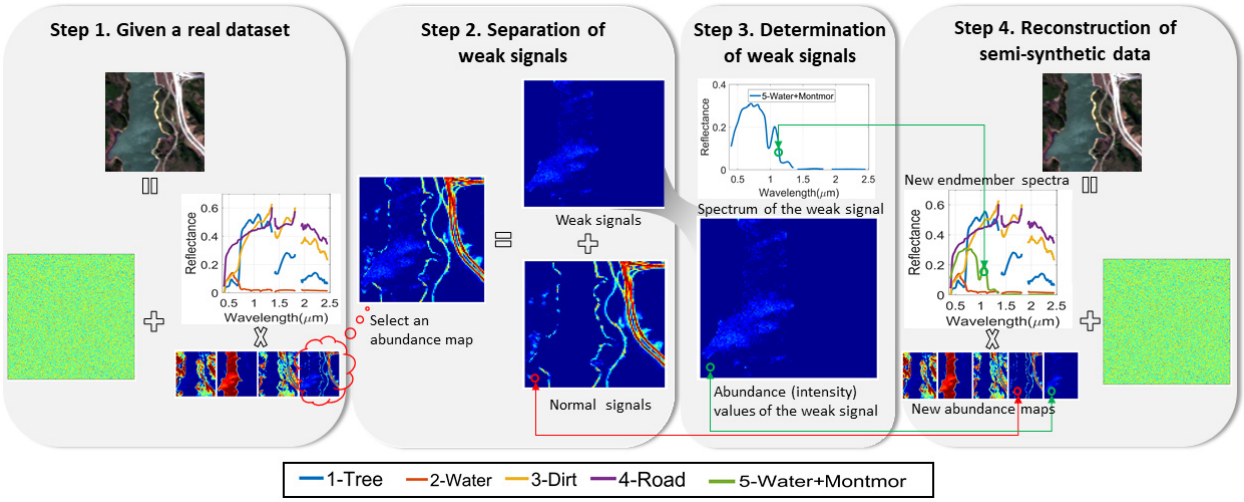



Fig. 2. Process of reconstructing a semi-synthetic real dataset containing weak signals. In the first step, we assume there exists a real hyperspectral image with endmember signatures and corresponding abundance fractions available. In the second step, we select an abundance map and define a region to extract the fractions. Therefore, the intensity values of weak signals can be determined based on abundance values in that region. In particular, in this figure, we observe that the abundance map corresponding to the *road* contains low fractions in the region of the *river*. We, therefore, extract abundance fractions from the region of the river in the abundance map corresponding to the road and treat them as the abundance map of a weak endmember. In the third step, we select an endmember signature corresponding to the weak endmember from the spectral library, e.g., the USGS library. In the final step, we reconstruct a semi-synthetic dataset containing newly added weak signals.

We first introduce the real hyperspectral datasets, which is known as *Jasper Ridge*. The widely used Jasper Ridge dataset contains 224 bands covering the wavelengths from 380 to 2500 nm, with a 10-nm spectral resolution and $512 \times 614$ pixels. A sub-image is selected from the original image with $100 \times 100$ pixels and 198 pruned bands (1–3, 108–112, 154–166, and 220–224 are removed due to the low-SNR and water-vapor absorption). There are four representative types of ground objects, i.e., *road*, *soil*, *water*, and *tree*.

Here, we introduce the process of generating semi-synthetic real hyperspectral data based on a linear mixing model. In terms of Jasper Ridge, to generate its weak signal version, we first assume that a hyperspectral image with endmember and abundance priors is known in advance. Then, we extract partial abundance fractions from a defined region of a given abundance map and treat it as an abundance map of weak endmember, denoted as $\zeta$. Next, we select a spectral signature from USGS as prior knowledge of weak endmember, denoted as $V$. Finally, we can reconstruct a semi-synthetic real hyperspectral image $\hat{Y}$ containing one weak ground object. The process can be expressed as

$$\hat{Y} = [W, V][H, \zeta] + G \tag{20}$$

where $W$ and $H$, respectively, represent the normal endmember and the corresponding abundance. $G = |Y - WH|$ denotes the noises. The process of generating a semi-synthetic hyperspectral image containing weak signals can be found in Fig. 2.

Based on the abovementioned reconstruction process, in this article, we provide two weak signal versions of Jasper Ridge. The first contains only one weak endmember called *Water + Montmor*, and its corresponding maximum abundance purity is 0.1. The second considers two weak endmembers, including *Water + Montmor* and *Sagebrush Sage-Leaves-1 dry* (called *Sagebrush* for short), and the maximum abundance purity of two signals equally is 0.1.

*5) Hyperspectral Real Datasets:* Cuprite is a benchmark real hyperspectral dataset for hyperspectral mixture spectral analysis, with data gathered by the Airborne Visible Infrared Imaging Spectrometer in Las Vegas, NV, USA. After removing noise and water absorption bands (the excluded bands were 1–6, 104–113, 148–167, and 220–224) and capturing a desired region of interest, we use a subarea of Cuprite sized $250 \times 191 \times 183$.

Nascimento and Dias [20] estimate 14 types of endmembers for unmixing purposes. Qian *et al.* [26] merely consider ten endmembers. Lu *et al.* [27] hold 12 endmembers for unmixing due to subtle differences among some spectra of the same mineral with different chemical compositions. In this manuscript, 12 endmembers are adopted for subsequent unmixing tasks, including *Alunite*, *Andradite*, *Buddingtonite*, *Dumortierite*, *Kaolinite#1*, *Kaolinite#2*, *Muscovite*, *Montmorillonite*, *Nontronite*, *Pyrope*, *Sphene*, and *Chalcedony*. Despite 12 normal endmembers, we select a spectrum *Kaolin/Smect KLF506 95%K* (or called *Smect* for short) from USGS library. The main reason to select that signature is that its corresponding abundances estimated by the nonnegative least-square method are subtle and structurally distributed, which can be treated as so-called weak endmembers.

### B. Experiments on Synthetic Data With Weak Signals

*1) Experiment 1 (Purity Analysis):* This experiment considers the impact of the purity (e.g., the mixing fraction) of the weak endmember on unmixing performances, where the purity is varied from 0.1 to 0.4, and one weak endmember is adopted in the synthetic image. Table III records the RMSE results of the weak endmember in different intensity situations.

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TABLE III

RMSE RESULTS OF THE WEAK ENDMEMBER WITH THE PURITY VARYING FROM 0.1 TO 0.4. THE BEST RESULTS ARE IN BOLD

| | HyperWeak | SGS-NMF | TV-RSNMF | SDNMF-TV | GL-NMF | $L_{1/2}$-NMF | MVC-NMF |
|---|---|---|---|---|---|---|---|
| 0.1 | **0.0241** | 0.0529 | 0.0455 | 0.0461 | 0.0425 | 0.0432 | 0.0397 |
| 0.2 | **0.0239** | 0.1069 | 0.0581 | 0.0584 | 0.0567 | 0.0583 | 0.0285 |
| 0.3 | **0.0217** | 0.1441 | 0.0456 | 0.0554 | 0.0483 | 0.0653 | 0.0283 |
| 0.4 | **0.0224** | 0.1918 | 0.0547 | 0.0503 | 0.0379 | 0.0456 | 0.0295 |

TABLE IV

SAD RESULTS OBTAINED FROM ALL ALGORITHMS CONDUCTED ON SYNTHETIC IMAGE WITH 30-dB NOISES. THE NUMBER OF WEAK ENDMEMBERS VARIES FROM ONE TO THREE. THE BEST RESULTS ARE IN BOLD

| | HyperWeak | SGS-NMF | TV-RSNMF | SDNMF-TV | GL-NMF | $L_{1/2}$-NMF | MVC-NMF |
|---|---|---|---|---|---|---|---|
| 1 | **0.0003** | 0.4866 | 0.0384 | 0.0266 | 0.1014 | 0.0604 | 0.0278 |
| 2 | **0.0001** | 0.4995 | 0.0623 | 0.0478 | 0.0660 | 0.0631 | 0.2287 |
| 3 | **0.0019** | 0.7216 | 0.0830 | 0.0806 | 0.0836 | 0.0814 | 0.1605 |

TABLE V

RMSE RESULTS OBTAINED FROM ALL ALGORITHMS CONDUCTED ON SYNTHETIC IMAGE WITH 30-dB NOISES, AND THE NUMBER OF WEAK ENDMEMBERS VARIES FROM ONE TO THREE. THE BEST RESULTS ARE IN BOLD

| | HyperWeak | SGS-NMF | TV-RSNMF | SDNMF-TV | GL-NMF | $L_{1/2}$-NMF | MVC-NMF |
|---|---|---|---|---|---|---|---|
| 1 | **0.0243** | 0.0658 | 0.0533 | 0.0444 | 0.0783 | 0.0622 | 0.0392 |
| 2 | **0.0414** | 0.0745 | 0.0458 | 0.0542 | 0.0422 | 0.0428 | 0.0613 |
| 3 | **0.0448** | 0.0767 | 0.0553 | 0.0568 | 0.0515 | 0.0517 | 0.0895 |

As shown in Table III, under entire scenarios, all NMF-based extensions have distinct results, where HyperWeak can produce the lowest RMSE results. Among comparison algorithms, MVC-NMF provides the suboptimal unmixing results, primarily because it reinforces a minimum simplex volume constraint to the endmembers, which relatively guarantees the unmixing performances of prior knowledge.

*2) Experiment 2 (Robustness Analysis):* This experiment validates the robustness of unmixing performances over the different numbers of weak endmembers. For the synthetic image, the intensity is fixed at 0.15, and the number of weak endmembers varies from 1 to 3. Tables IV and V, respectively, detail the SAD and RMSE results over different numbers of weak endmembers. Normally, when the number of weak endmember goes up, the unmixing task may be more difficult. For HyperWeak, for the different numbers of weak endmember, it relatively provides better results than those of counterparts.

*3) Experiment 3 (Noise Analysis):* To verify the impact of different noise levels on unmixing performances, this experiment considers a synthetic image with one weak endmember of 0.15 intensity, five normal endmembers of 0.8 maximal purity, and multiple SNRs varying from 10 to 40 dB. Fig. 3(a) and (b), respectively, demonstrates mSAD and mRMSE results under different SNR situations. From Fig. 3(a) and (b), we find that when the noise is poor, such as 10 dB, the NMF-based methods produce relatively bad results. For 20, 30, and 40 dB, HyperWeak provides the best unmixing performances than those of competitive methods. Fig. 3(c) and (d), respectively, displays the SAD and RMSE results of the weak endmember in different noise levels.

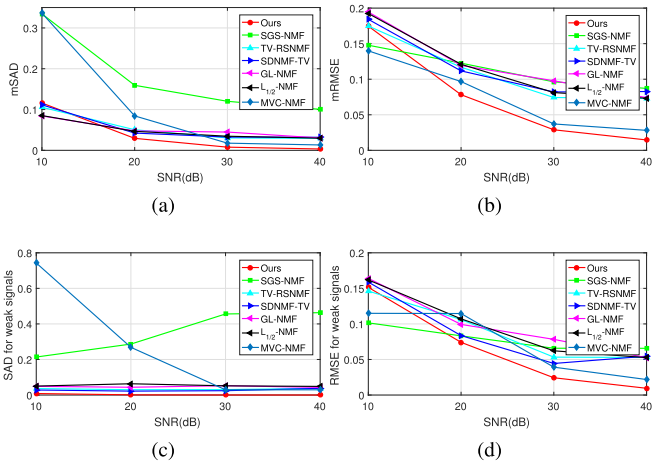



Fig. 3. Impact of noises to the unmixing results. (a) mSAD versus SNRs. (b) mRMSE versus SNRs. (c) SAD values of weak endmembers versus SNRs. (c) RMSE values of weak endmembers versus SNRs.

Although SAD results obtained from HyperWeak are the lowest than those of competitive methods in all noise situations, the RMSE results in 10 dB may be disappointing mainly because of bad noise interference. Besides, two estimated abundance maps obtained from entire methods are displayed in Fig. 4(b) and (c), respectively. The maximum abundance fraction of normal endmembers is fixed at 0.8, while that of weak endmember is fixed at 0.15. Fig. 4(a) represents the corresponding ground truth. The RMSE results are supplied to the bottom of each abundance map. In Fig. 4(b), all methods can relatively estimate accurate abundance fractions even the noise is poor, but it is not consistently found in Fig. 4(c), in which several methods not only generate inaccurate maps in 10 dB or even 20-dB scenarios, but also show unmixing errors in some local areas. Among comparative algorithms, HyperWeak has the ability to estimate abundance maps for the 20-dB situation.

*4) Experiment 4 (Accuracy Analysis):* To provide quantitative and comparative results of seven methods in the synthetic dataset, this experiment was conducted on a synthetic image with one weak endmember of 0.15 purity, five normal endmembers of 0.8 maximal purity, and 30-dB noises. The overall results are tabulated in Table VI, where HyperWeak shows better unmixing performances particularly for endmembers #2, #3, #4, mSAD, and mRMSE than those of other counterparts. Besides, in terms of the weak endmember (endmember #6), HyperWeak has the lowest degradation of prior knowledge and the best estimation for abundance maps. For the comparison algorithms, MVC-NMF shows relatively better results for endmembers #1 and #5, because it involves a minimum volume-based constraint that can estimate desirable endmembers.

### C. Experiments on Semi-Synthetic Real Datasets With Weak Signals

*1) Experiment 5 (Validations on Jasper Ridge With One Weak Endmember Prior Knowledge):* To validate the experimental performances of different methods in a real situation manner, this experiment was conducted in the Jasper Ridge dataset. Under our previous reconstruction process, Jasper Ridge is reconstructed with five ground objects composed of

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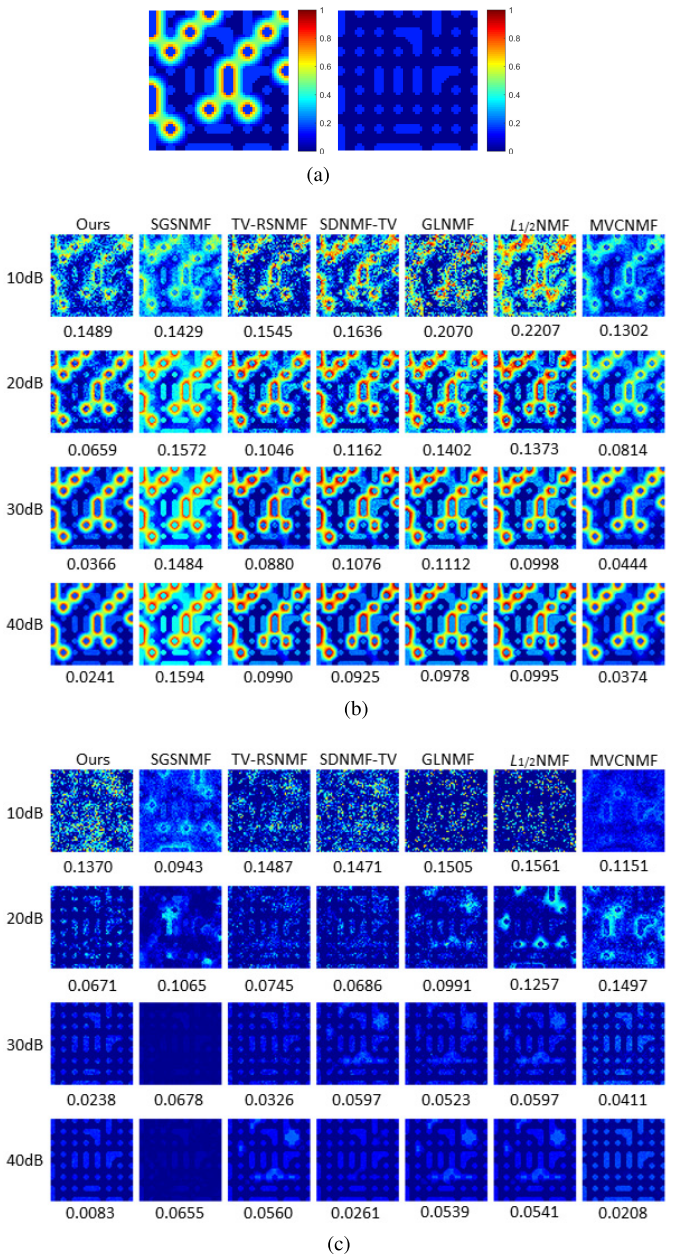



Fig. 4. Illustration of estimated abundance maps obtained from different methods over noisy data. From leftmost column to rightmost column, the abundance maps are generated by HyperWeak, SGS-NMF, TV-RSNMF, SDNMF-TV, GLNMF, $L_{1/2}$-NMF, and MVC-NMF. From top row to bottom row, the abundance maps are generated under 10, 20, 30, and 40 dB. (a) Two reference maps for (b) and (c). The quantitative RMSE results are displayed below each abundance map. (a) Ground-truth (GT). (b) Abundance maps obtained from the normal signals. (c) Abundance maps obtained from the weak endmembers.

TABLE VI

OVERALL RESULTS OBTAINED FROM ALL ALGORITHMS CONDUCTED ON SYNTHETIC IMAGE WITH 30-dB NOISES, FIVE NORMAL ENDMEMBERS, AND ONE WEAK ENDMEMBER. THE BEST RESULTS ARE IN BOLD, AND THE SUBOPTIMAL VALUES ARE UNDERLINED

| | HyperWeak | SGS-NMF | TV-RSNMF | SDNMF-TV | GL-NMF | $L_{1/2}$-NMF | MVC-NMF |
|---|---|---|---|---|---|---|---|
| Endmember #1 | 0.0094 | 0.0109 | 0.0258 | 0.0274 | 0.0265 | 0.0273 | **0.0041** |
| Endmember #2 | **0.0073** | 0.0534 | 0.0293 | 0.0382 | 0.0434 | 0.0298 | 0.0085 |
| Endmember #3 | **0.0043** | 0.0375 | 0.0278 | 0.0371 | 0.0277 | 0.0266 | 0.0171 |
| Endmember #4 | **0.0073** | 0.0813 | 0.0286 | 0.0248 | 0.0324 | 0.0315 | 0.0347 |
| Endmember #5 | 0.0182 | 0.0515 | 0.0322 | 0.0416 | 0.0375 | 0.0319 | **0.0134** |
| Endmember #6 | **0.0003** | 0.4866 | 0.0384 | 0.0266 | 0.1014 | 0.0604 | 0.0278 |
| mSAD | **0.0078** | 0.1202 | 0.0304 | 0.0326 | 0.0448 | 0.0346 | 0.0176 |
| mRMSE | **0.0289** | 0.0963 | 0.0743 | 0.0821 | 0.0978 | 0.0812 | 0.0371 |
| mRMSE for Weak Endmembers | **0.0243** | 0.0658 | 0.0533 | 0.0444 | 0.0783 | 0.0622 | 0.0392 |

TABLE VII

OVERALL RESULTS OBTAINED FROM ALL ALGORITHMS CONDUCTED ON JASPER RIDGE WITH FOUR REAL LAND COVERS AND ONE WEAK OBJECT (WATER + MONTMOR). THE BEST RESULTS ARE IN BOLD, AND THE SUBOPTIMAL VALUES ARE UNDERLINED

| Algorithms | HyperWeak | SGS-NMF | TV-RSNMF | SDNMF-TV | GL-NMF | $L_{1/2}$-NMF | MVC-NMF |
|---|---|---|---|---|---|---|---|
| Tree | **0.1353** | 0.1644 | 0.1599 | 0.1462 | 0.1544 | 0.1541 | 0.1875 |
| Water | 0.1069 | 0.1215 | 0.1410 | 0.1463 | 0.1013 | **0.1012** | 0.1792 |
| Dirt | **0.0842** | 0.1016 | 0.1124 | 0.0971 | 0.1097 | 0.1090 | 0.0888 |
| Road | 0.0550 | 0.0695 | 0.0513 | **0.0487** | 0.0496 | 0.0500 | 0.0543 |
| Water+ Montmor | **0.0107** | 0.7485 | 0.2327 | 0.3962 | 0.2619 | 0.2615 | 0.4386 |
| mSAD | **0.0784** | 0.2411 | 0.1395 | 0.1669 | 0.1354 | 0.1352 | 0.1897 |
| mRMSE | **0.0952** | 0.1068 | 0.1029 | 0.1013 | 0.1037 | 0.1040 | 0.1182 |
| RMSE for Weak Endmembers | **0.0738** | 0.0984 | 0.0842 | 0.0829 | 0.0874 | 0.0880 | 0.1411 |

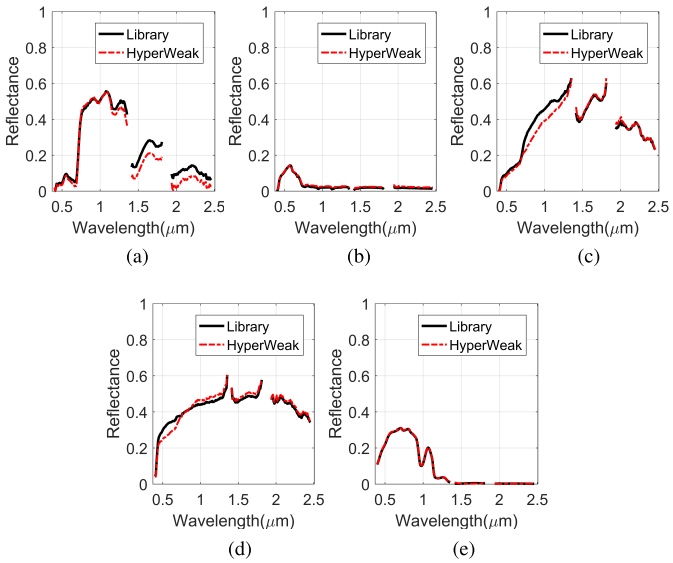



Fig. 5. Visual comparisons of library and estimated signatures. (a) Tree. (b) Water. (c) Dirt. (d) Road. (e) Water + Montmor.

one weak endmember and four normal endmembers. Table VII tabulates overall unmixing results in detail, where HyperWeak shows better endmember estimation performances for *Tree* and *Dirt* and simultaneously ensures low deterioration of *a priori Water + Montmor*. Besides, HyperWeak also provides lower mSAD and mRMSE results for prior knowledge than those of competitive methods. Among considered methods, $L_{1/2}$-NMF and SDNMF-TV can, respectively, produce accurate estimation for *Water* and *Road*. Fig. 5 visually displays the spectral curves obtained from the library and signatures estimated by HyperWeak. Fig. 6 compares the differences between ground truth and estimated abundance maps.

*2) Experiment 6 (Validations on Jasper Ridge With Two Weak Endmember Prior Knowledge):* This experiment validates the performances of different algorithms on Jasper datasets with two man-made weak objects. Unlike experiment 5, we further supply another ground object called *Sagebrush*. Overall experimental results are tabulated in Table VIII in detail. In this table, HyperWeak provides the best estimations for two normal land covers, i.e., *Water* and *Road*, and equally has the lowest spectral variations for two

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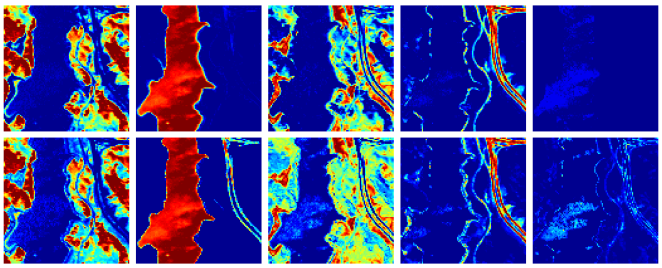


Fig. 6. Visual comparisons of ground truth (first row) and estimated abundance fractions (second row). From leftmost column to rightmost column, the five abundance maps, respectively, represent *Tree*, *Water*, *Dirt*, *Road*, and *Water + Montmor*.

TABLE VIII

OVERALL RESULTS OBTAINED FROM ALL ALGORITHMS CONDUCTED ON JASPER RIDGE WITH FOUR REAL LAND COVERS AND TWO WEAK OBJECTS (SAGEBRUSH AND WATER + MONTMOR). THE BEST RESULTS ARE IN BOLD, AND THE SUBOPTIMAL VALUES ARE UNDERLINED

| Algorithms | HyperWeak | SGS-NMF | TV-RSNMF | SDNMF-TV | GL-NMF | $L_{1/2}$-NMF | MVC-NMF |
|---|---|---|---|---|---|---|---|
| Tree | 0.1104 | 0.1585 | 0.1224 | 0.1173 | **0.1061** | 0.1064 | 0.2203 |
| Water | **0.1094** | 0.1145 | 0.1547 | 0.1710 | 0.1203 | 0.1201 | 0.3023 |
| Dirt | 0.0217 | 0.1294 | 0.0357 | 0.0365 | 0.0213 | **0.0211** | 0.1279 |
| Road | **0.0472** | 0.0607 | 0.0526 | 0.0512 | 0.0530 | 0.0534 | 0.0704 |
| Sagebrush | **0.0054** | 0.7894 | 0.1859 | 0.1864 | 0.2169 | 0.2178 | 0.4277 |
| Water+ Montmor | **0.0047** | 0.5977 | 0.2472 | 0.3481 | 0.2898 | 0.2883 | 0.3609 |
| mSAD | **0.0498** | 0.3084 | 0.1331 | 0.1517 | 0.1346 | 0.1345 | 0.2516 |
| mRMSE | **0.0909** | 0.1081 | 0.1013 | 0.1034 | 0.0990 | 0.0999 | 0.1226 |
| RMSE for Weak Endmembers | **0.0671** | 0.0739 | 0.0748 | 0.0763 | 0.0704 | 0.0727 | 0.1186 |

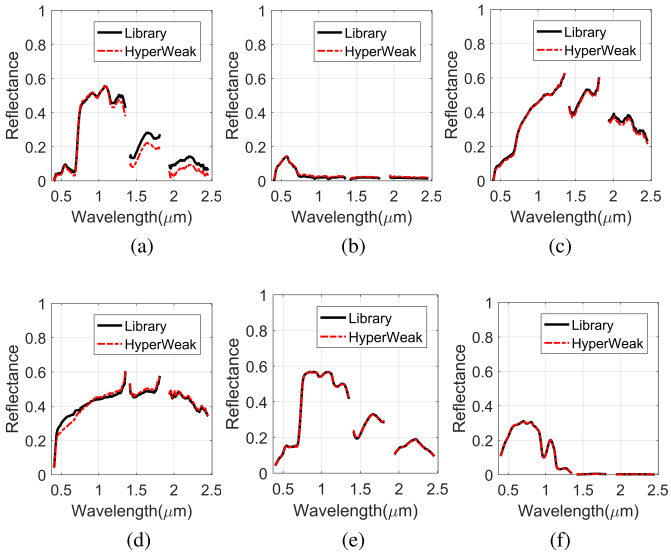



Fig. 7. Visual comparisons of library and estimated signatures. (a) Tree. (b) Water. (c) Dirt. (d) Road. (e) Sagebrush. (f) Water + Montmor.

weak endmembers. The lowest mSAD and mRMSE values prove the effectiveness of HyperWeak on Jasper Ridge that contains two weak endmembers. The visualization of estimated endmembers and abundance fractions is displayed in Figs. 7 and 8, respectively.

### D. Experiment on the Cuprite Real Dataset (Experiment 7)

This experiment is conducted on the Cuprite dataset by considering 12 dominated minerals and one weak mineral. Table IX tabulates overall unmixing results obtained from

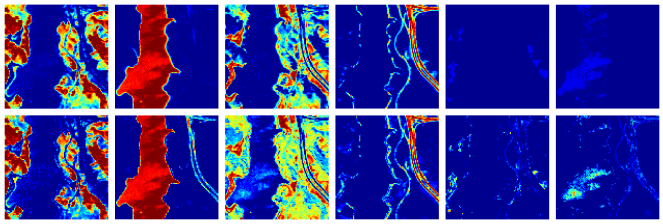


Fig. 8. Visual comparisons of ground truth (first row) and estimated abundance fractions (second row). From leftmost column to rightmost column, the six abundance maps, respectively, represent *Tree*, *Water*, *Dirt*, *Road*, *Sagebrush*, and *Water + Montmor*.

TABLE IX

OVERALL RESULTS OBTAINED FROM ALL ALGORITHMS CONDUCTED ON CUPRITE DATASET WITH 12 MINERALS AND ONE WEAK OBJECT (SMECT). THE BEST RESULTS ARE IN BOLD, AND THE SUBOPTIMAL VALUES ARE UNDERLINED

| Algorithms | HyperWeak | SGS-NMF | TV-RSNMF | SDNMF-TV | GL-NMF | $L_{1/2}$-NMF | MVC-NMF |
|---|---|---|---|---|---|---|---|
| Alunite | 0.0974 | 0.1005 | **0.0956** | 0.0959 | **0.0956** | 0.0958 | 0.1182 |
| Andradite | **0.1068** | 0.1159 | 0.1116 | 0.1116 | 0.1119 | 0.1113 | 0.1232 |
| Buddingtonite | 0.0641 | 0.0651 | 0.0632 | 0.0638 | **0.0622** | 0.0630 | 0.0712 |
| Dumortierite | **0.0802** | 0.0840 | 0.0831 | 0.0814 | 0.0838 | 0.0838 | 0.1384 |
| Kaolinite#1 | **0.0888** | 0.0890 | 0.0907 | 0.0904 | 0.0907 | 0.0909 | 0.1227 |
| Kaolinite#2 | 0.0570 | **0.0540** | 0.0592 | 0.0589 | 0.0589 | 0.0586 | 0.0631 |
| Muscovite | **0.0843** | 0.0862 | 0.0844 | 0.0845 | 0.0844 | 0.0844 | 0.0880 |
| Montmorillonite | **0.0626** | 0.0655 | 0.0646 | 0.0642 | 0.0644 | 0.0645 | 0.0729 |
| Nontronite | 0.0710 | 0.0691 | **0.0688** | 0.0692 | 0.0689 | **0.0688** | 0.0763 |
| Pyrope | 0.1286 | 0.0758 | 0.0911 | 0.0887 | 0.0913 | 0.0911 | **0.0575** |
| Sphene | 0.0639 | 0.0900 | 0.0852 | 0.0865 | 0.0837 | 0.0838 | **0.0568** |
| Chalcedony | 0.1300 | **0.1245** | 0.1266 | 0.1254 | 0.1260 | 0.1262 | 0.1281 |
| Smect | **0.0011** | 0.0715 | 0.1238 | 0.1033 | 0.1314 | 0.1261 | 0.0699 |
| mSAD | 0.0797 | 0.0839 | 0.0883 | 0.0865 | 0.0887 | 0.0883 | 0.0913 |

entire algorithms. According to Table IX, we observe that HyperWeak shows compelling unmixing performances for five mineral signatures with the lowest degradation of prior knowledge and the best mSAD results. The competitive methods, including SGS-NMF, TV-RSNMF, GL-NMF, and MVC-NMF, equally estimate the best results for two minerals. Visualizations of endmember signatures and corresponding estimated abundance maps are displayed in Figs. 9 and 10, respectively. As there is no publicly available ground truth of abundance maps, the nonnegative least-square method is used to unmix the fractional abundances of *Smect* for reference purposes. The fractional abundances of *Smect* estimated by HyperWeak are observed, which show correlations to that estimated by the nonnegative least-square method.

### E. Experiments on Model Robustness

*1) Experiment 8 (Parameter Analysis):* In this article, the proposed HyperWeak requires two regularization parameters $\tau$ and $\lambda$, where the former strictly affects the importance of the *a priori*, and the latter impacts the sparseness of abundance matrix. To validate the impacts of two parameters to HyperWeak, this experiment was conducted on a synthetic dataset (see Table II for parameter details). For the purpose of observing unmixing accuracy changes, $\lambda$ is varied in $\{1e-7, 5e-7, 1e-6, 5e-6, 1e-5, 5e-5, 1e-4, 5e-4, 1e-3, 5e-3, 1e-2, 5e-2, 1e-1\}$, and $\tau$ is varied in $\{1, 10, 50, 100, 300, 500, 700, 1000\}$. As shown in Fig. 11(a) and (b), when tuning $\tau$, SAD and RMSE values show a similar change tendency. Here, $\tau$ is fixed at 500 for subsequent experiments. For $\lambda$, the best result is found when we set $\lambda$ to $1e-3$ (see Fig. 12).

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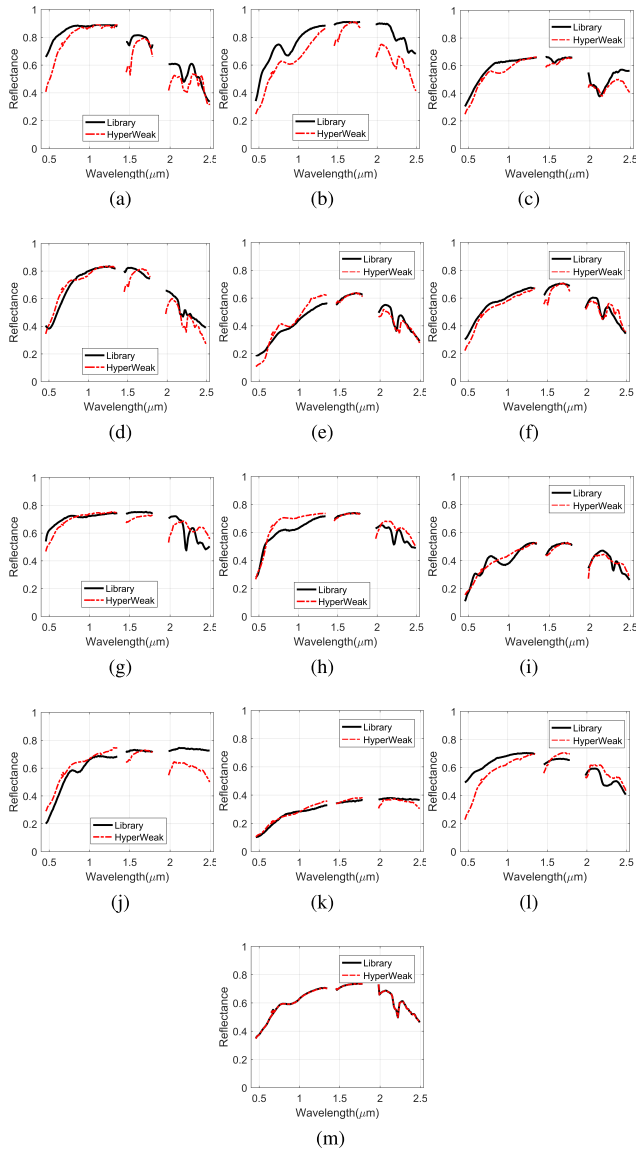



Fig. 9. Visual comparisons of library and estimated signatures. (a) Alunite. (b) Andradite. (c) Buddingtonite. (d) Dumortierite. (e) Kaolinite#1. (f) Kaolinite#2. (g) Muscovite. (h) Montmorillonite. (i) Nontronite. (j) Pyrope. (k) Sphene. (l) Chalcedony. (m) Smect.

*2) Experiment 9 (Initialization Analysis):* To investigate the influence of initialization, this article proposes a new method called Robust-OSP and conducts an experiment to verify the performances of considered initialization methods, including randomized method, VCA, and Robust-OSP. This experiment was conducted on a synthetic dataset where the initialization methods were independently performed ten times. Table X displays mSAD, mRMSE, and corresponding standard deviation results obtained from three methods. Under ten independent runs, Robust-OSP shows the most stable results, indicating that the unmixing results obtained from Robust-OSP-based initialization can be repeatable. For the randomized method, it triggers poor validation values. We are also aware that VCA is a widely used initialization strategy for NMF-based variants owing to its light computational burden and relatively better initialization performances. However, it requires a projection process related to randomly generated vectors, resulting in its unrepeatable manner.

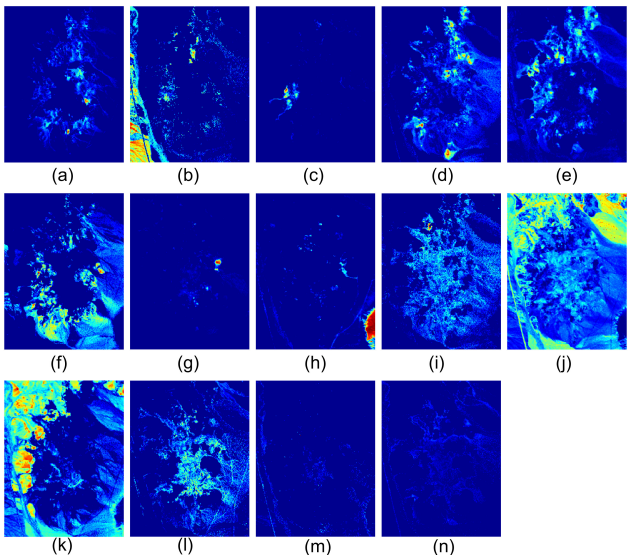



Fig. 10. Visualization of estimated abundances. (a)–(m) *Alunite*, *Andradite*, *Buddingtonite*, *Dumortierite*, *Kaolinite#1*, *Kaolinite#2*, *Muscovite*, *Montmorillonite*, *Nontronite*, *Pyrope*, *Sphene*, *Chalcedony*, and *Smect*, respectively. (n) Abundance map *Smect* estimated by the nonnegative least-square method.

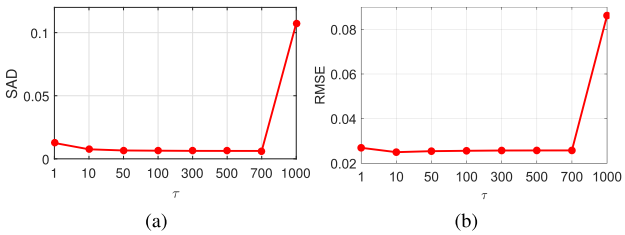



Fig. 11. (a) SAD accuracy with varying regularization parameter $\tau$ and (b) RMSE values for the weak endmember over different $\tau$ values.

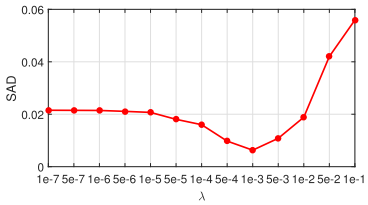



Fig. 12. SAD accuracy with varying regularization parameter $\lambda$.

TABLE X
OVERALL RESULTS OBTAINED FROM THREE INITIALIZATION METHODS UNDER TEN INDEPENDENT RUNS. THE STANDARD DEVIATION (Std.) ARE DISPLAYED IN PARENTHESES. THE BEST RESULTS ARE IN BOLD

| | Randomized init. | VCA | Robust-OSP |
|---|---|---|---|
| mSAD(std.) | 0.3578(0.0230) | 0.0089(0.0070) | **0.0048(0)** |
| mRMSE(std.) | 0.1719(0.0160) | 0.0447(0.0113) | **0.0295(0)** |
| mRMSE for weak endmembers(std.) | 0.1862(0.0050) | 0.0691(0.0289) | **0.0466(0)** |

*3) Experiment 10 (Convergence Analysis):* To observe the convergence performances of HyperWeak, this experiment visually illustrates the process of optimization.

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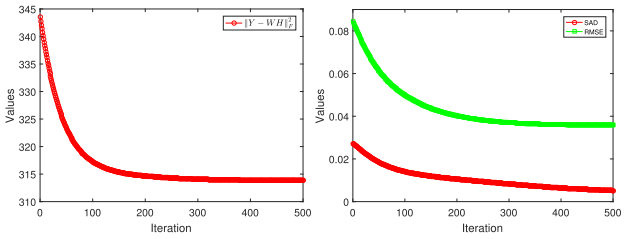



Fig. 13. Convergence curves under 500 iterations. (a) Objective values versus iterations. (b) SAD and RMSE values versus iteration.

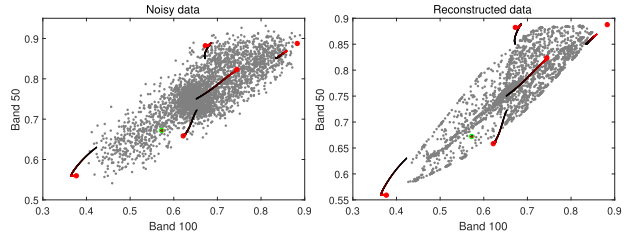



Fig. 14. Visual display of optimization tracking. The gray point, red point, green point, and black-red gradient line, respectively, denote the hyperspectral data, real endmember, weak endmember, and optimization direction. (a) Optimization directions in the noisy data. (b) Optimization directions in the reconstructed data.

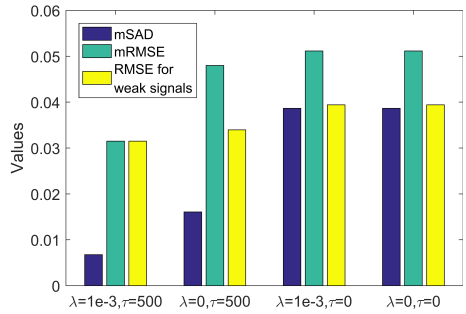



Fig. 15. Experimental performances with different regularization conditions.

Fig. 13(a) and (b), respectively, demonstrates the tendency of objective values and quantitative results. As shown in Fig. 13(a), HyperWeak gets a fast descent behavior under the first 100 iterations. Similarly, the quantitative values tend to be low when optimizing HyperWeak and get relatively stable after 300 iterations [Fig. 13(b)]. To further visually demonstrate the optimization process, Fig. 14(a) records the optimization steps on noisy data. From Fig. 14(a), we find that the prior knowledge (green point) is negligibly optimized due to the specially designed regularization term. Compared with Fig. 14(a), Fig. 14(b) indicates that HyperWeak well mitigates the interference of noise and approximately reconstructs original data simplex.

*4) Experiment 11 (Regularization Analysis):* This experiment verifies the impact of different regularization terms on the experimental performances. Fig. 15 displays the experimental values regarding mSAD, mRMSE, and RMSE (for weak endmembers) over different regularization combinations. If $\lambda$ is fixed at 0, it means that the corresponding sparsity constraint is not considered, and similarly, $\tau = 0$ denotes that the supervised part is ignored. When $\tau$ is fixed at 0, its RMSE is higher than that is set to 500, which implies that the utilization of weak signals works in the proposed method. Similarly, when $\lambda$ is fixed at 0, the overall results are not good as the situation where $\lambda$ is set to $1e-3$. According to Fig. 15, when two regularization terms are simultaneously considered, the proposed method yields the best results, followed by the combination ($\lambda = 0$ and $\tau = 500$) that ensures suboptimal values owing to its constraint to the prior knowledge.

TABLE XI

ABLATION EXPERIMENTS CONDUCTED ON TWO VERSIONS OF JASPER RIDGE DATASET

| | Sparsity ($\lambda$) | Prior Knowledge ($\tau$ ) | mSAD | mRMSE |
|---|---|---|---|---|
| Jasper Ridge #1 | × | × | 0.1560 | 0.1032 |
| | ✓ | × | 0.1475 | 0.1031 |
| | ✓ | ✓ | 0.0799 | 0.1027 |
| Jasper Ridge #2 | × | × | 0.1308 | 0.0939 |
| | ✓ | × | 0.1367 | 0.0939 |
| | ✓ | ✓ | 0.0736 | 0.0933 |

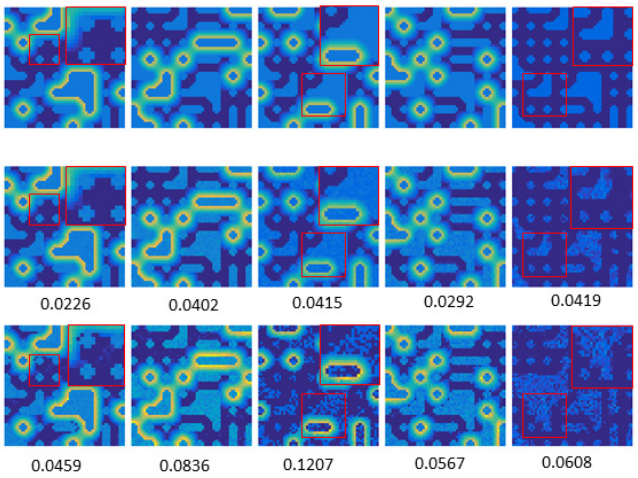



Fig. 16. Visual comparisons among ground truth (first row), our reweighting strategy (second row), and traditional strategy (bottom row). The RMSE results are displayed below the abundance maps. Three regions of interest are zoomed for better observations.

The overall ablation results obtained on Jasper Ridge datasets are tabulated in Table XI, where Jasper Ridge #1 and #2, respectively, represent the dataset that contains one weak endmember and two endmembers, As shown in Table XI, for two versions of Jasper Ridge datasets, when simultaneously considering two regularizers, the proposed method ensures the best unmixing results, which outperforms the case that merely utilizes sparsity constraint. The details and experiments have been added to the revised manuscript.

For the proposed method, we develop an improved reweighting strategy that first smooths optimized abundance maps for the sake of preserving local structures and then generates a reweighting matrix. To verify the performances of the proposed strategy, Fig. 16 provides the comparisons of the visual and quantitative results in three regions. Compared with the ground truth, our proposed reweighting method can provide better quantitative results and local recovery performances in three regions than the traditional reweighting strategy that $R$ is generated by $1 \oslash (|H| + \text{eps})$.

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## V. Discussion

One of the main purposes of these experiments is to verify the unmixing ability of the proposed method in a weak signal behavior. The observation to emerge from the result comparison is that the proposed method extends the discriminative representations in weak signal analysis for hyperspectral unmixing.

Hyperspectral unmixing generally relates to the process of distinguishing representative land covers from a subpixel viewpoint, which can be summarized as the procedure of identifying endmember and abundance parts. However, in real applications, observed spectral signatures may be mixed by the endmembers of minor objects, e.g., in the case of algae bloom, heavy metal pollution, and rare mineral exploration, which normally have subtle mixing fractions. This fact limits the accurate unmixing performance of traditional methods. Considering another truth that sparse regression-based methods require an independent library and NMF-based methods deteriorate prior knowledge, this article links two unmixing strategies and proposes an efficient unmixing model to handle weak signal situations. The major difference between the proposed method and other counterparts is that we emphasize an augmented endmember matrix that contains the endmembers of representative land covers and that of weak signals, i.e., prior knowledge, where prior knowledge is ensured by a fidelity term. As expected, our experiments demonstrate that the proposed method well controls the trade-off between unmixing performances and the abilities of weak signal separation.

We are aware the flaw of this article is that there may exist other potential reasons resulting in weak signal issues. The first is that when weak signals exist, their spectral signatures may also be highly correlated with those of other major land covers. This reason may be similar to the active research area of endmember variability. Another may relate to the low SNRs, i.e., some weak signals are generated owing to complex noises. In addition, in terms of the proposed method, the determination of the number of weak endmembers and the acquisition of pure signal library corresponding to weak endmembers may be difficult, where the former barely produces reliable estimations, and the latter is costly. To the best of our knowledge, the process of identifying weak endmember may require an image-based analysis or a ground survey process. These covered limitations will form our main future lines of research.

As mentioned in Section I, no one appears to consider the weak signal case during unmixing tasks, and yet, this topic deserves further exploitation, especially in a more complex scene, such as water monitoring, mineral exploitation, and so on. To solve this case, this article attempts to propose an efficient model to analyze weak signals, which obtains positive experimental results compared with some traditional unmixing methods.

## VI. Conclusion

This article investigates a special case of hyperspectral unmixing, which we assume there exist weak signals in the image. To solve this issue, we propose the HyperWeak model that contains two main parts, where the unsupervised part simultaneously estimates abundance and endmember matrices, and the supervised part ensures the low degradation of prior knowledge. To boost the robustness of HyperWeak, we further add a reweighted sparsity constraint, Nesterov's OGM, and a new endmember initialization method named Robust-OSP. Based on SAD and RMSE validation metrics, experiments conducted on synthetic and real hyperspectral images indicate that HyperWeak can improve the unmixing performances under weak signal scenarios.

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